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Normal distribution

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6-1 Normal Distributions

If a random variable has a probability distribution whose graph is continuous, bell-shaped, and symmetric, it is called a **normal distribution**. The graph is called a *normal distribution curve*.

The mathematical equation for a normal distribution is

$$y = \frac{e^{-(X-\mu)^2/(2\sigma^2)}}{\sigma \sqrt{2\pi}}$$

where $e \approx 2.718$ ($\approx$ means “is approximately equal to”)

$\pi \approx 3.14$

μ = population mean

σ = population standard deviation

The normal distributions are closely associated with many things such as:

Marks scored on the test

Heights of different persons

Size of objects produced by the machine

Blood pressure and so on.

Example 1: Find the probability density function of the normal distribution of the following data. $x = 2$, $\mu = 3$ and $\sigma = 4$.

Solution:

Given,

- *Variable (x) = 2*
- *Mean = 3*
- *Standard Deviation = 4*

Using formula of probability density of normal distribution

$$f(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Simplifying,

$$f(2, 3, 4) = 0.09666703$$

1. Bell-shaped curve

The graph looks like a smooth hill — highest in the middle and lower on both sides.

2. Mean = Median = Mode

The average value, the middle value, and the most common value are all the same and lie right at the center of the curve.

3. Unimodal (one peak)

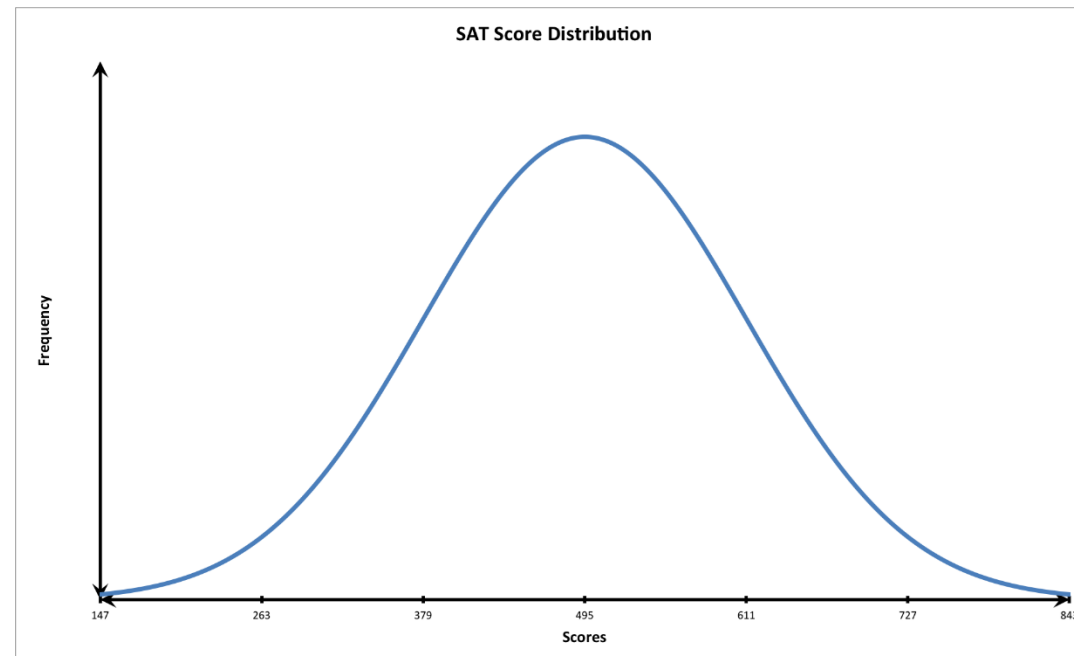
There is only **one** highest point — the curve has just one “top.”

4. Symmetric around the mean

If you draw a vertical line through the center, the left and right sides look exactly the same — like a mirror image.

5. Continuous curve

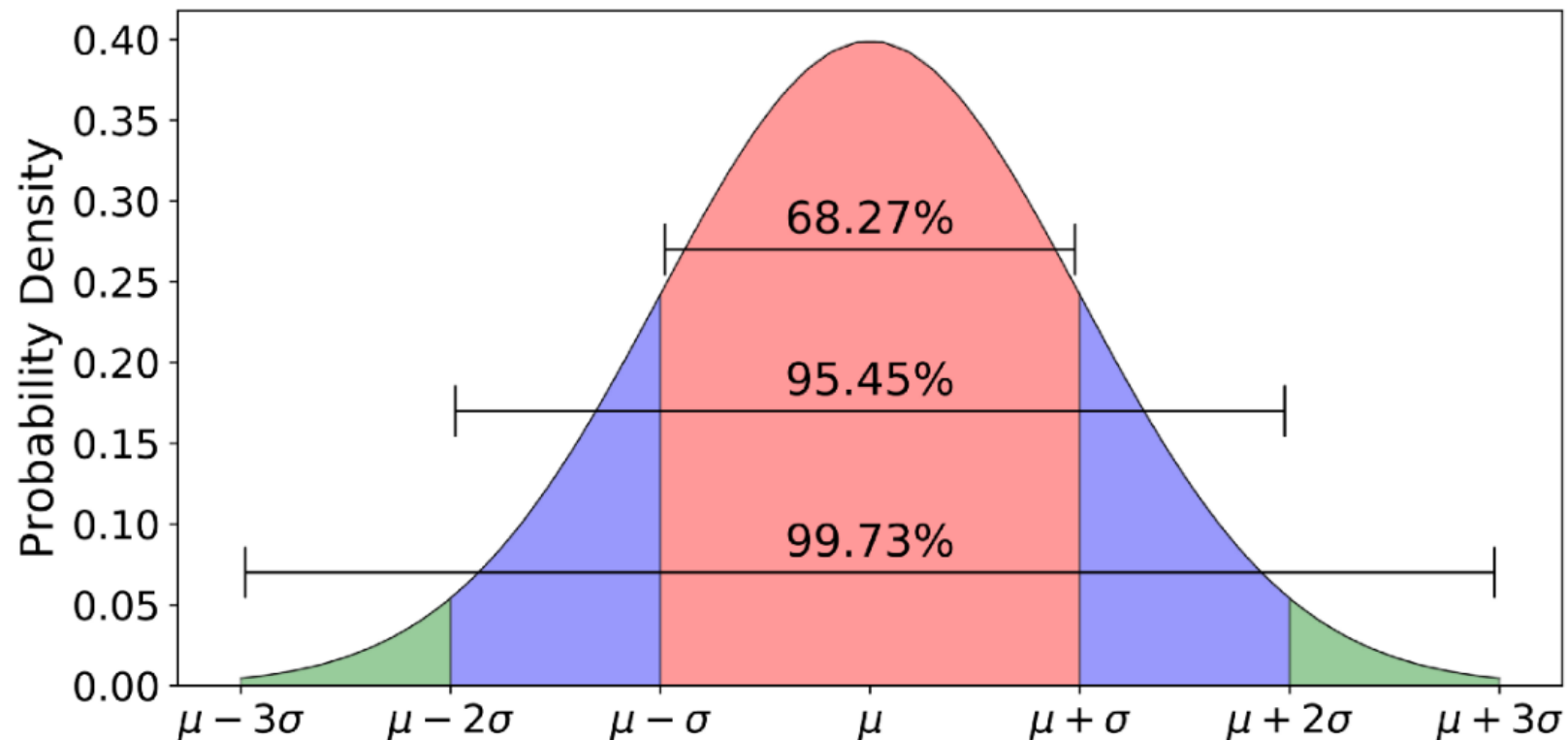
The line has **no breaks or gaps**. Every possible value of X has a corresponding height (Y value) on the curve.



Empirical rule

The **empirical rule**, or the 68-95-99.7 rule, tells you where most of your values lie in a normal distribution:

- ❑ Around 68% of values are within 1 standard deviation from the mean.
- ❑ Around 95% of values are within 2 standard deviations from the

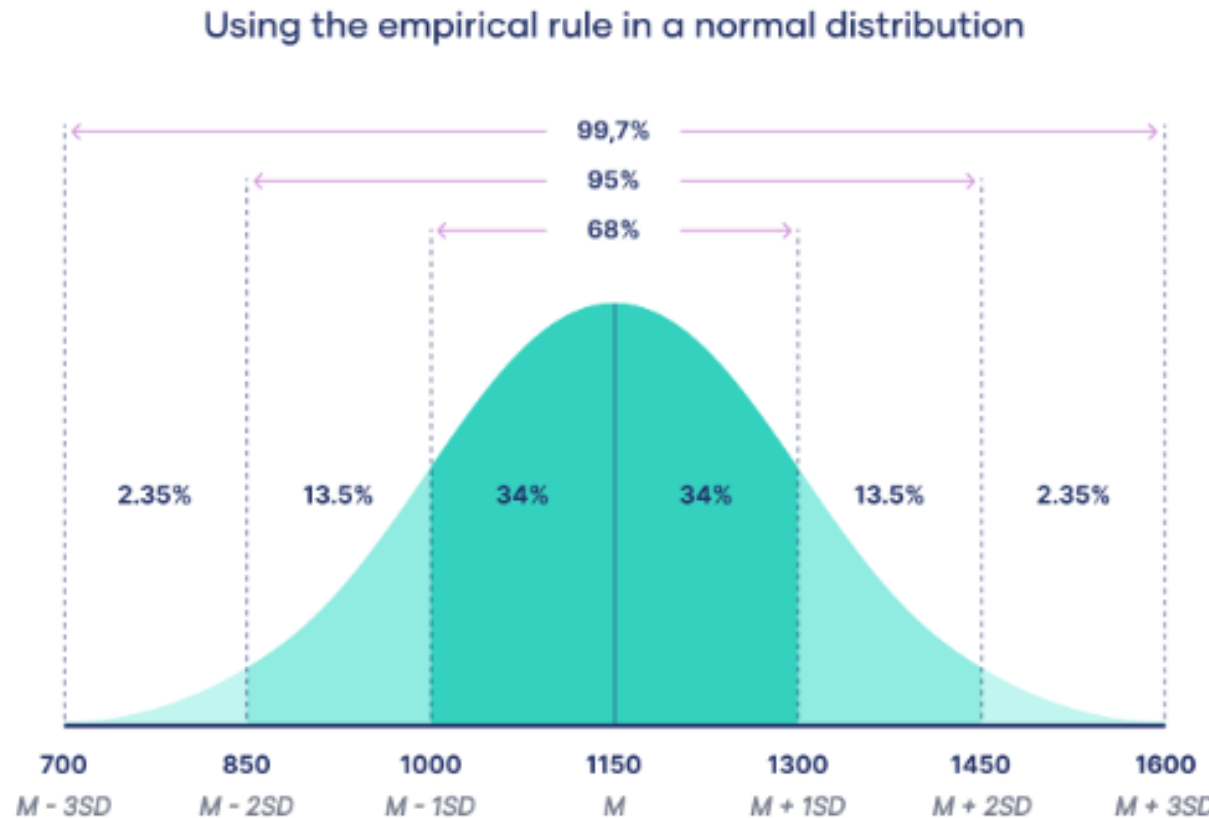


Example: Using the empirical rule in a normal distribution

You collect SAT scores from students in a new test preparation course. The data follows a normal distribution with a mean score (M) of 1150 and a standard deviation (SD) of 150.

Solution:

- ❑ Around 68% of scores are between 1,000 and 1,300, 1 standard deviation above and below the mean.
- ❑ Around 95% of scores are between 850 and 1,450, 2 standard deviations above and below the mean.
- ❑ Around 99.7% of scores are between 700 and 1,600, 3 standard deviations above and below the mean.



The Standard Normal Distribution

The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1.

Finding Areas Under the Standard Normal Distribution Curve

EXAMPLE 6-1

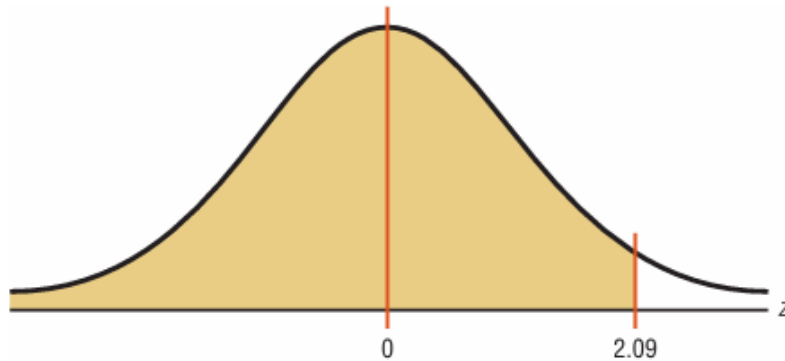
Find the area under the standard normal distribution curve to the left of $z = 2.09$.

SOLUTION

Step 1 Draw the figure. The desired area is shown in Figure 6-8.

FIGURE 6-8

Area Under the Standard
Normal Distribution Curve for
Example 6-1



Step 2 We are looking for the area under the standard normal distribution curve to the left of $z = 2.09$. Since this is an example of the first case, look up the area in the table. It is 0.9817. Hence, 98.17% of the area is to the left of $z = 2.09$.

EXAMPLE 6-2

Find the area under the standard normal distribution curve to the right of $z = -1.14$.

EXAMPLE 6-2

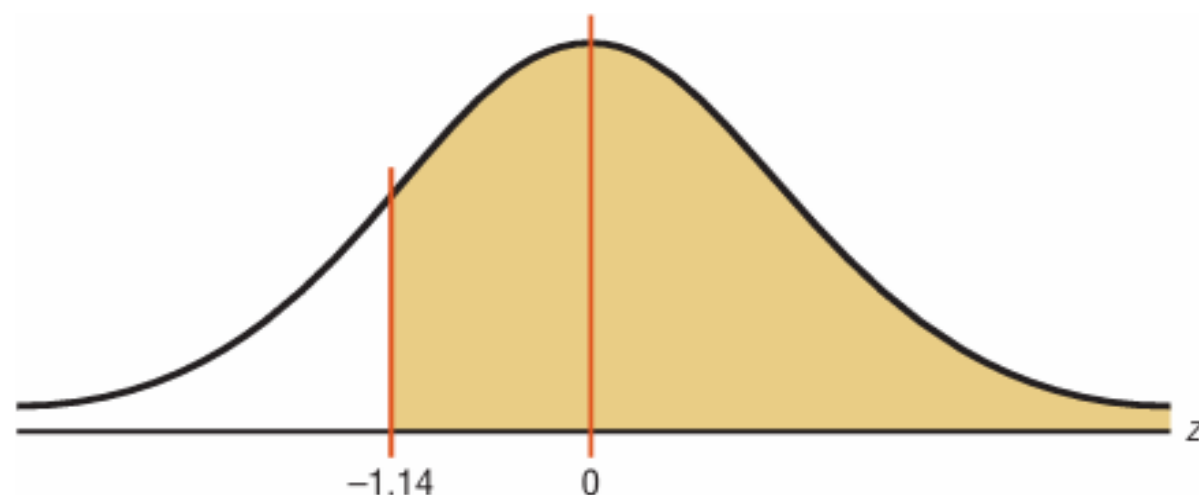
Find the area under the standard normal distribution curve to the right of $z = -1.14$.

SOLUTION

Step 1 Draw the figure. The desired area is shown in Figure 6-9.

FIGURE 6-9

Area Under the Standard
Normal Distribution Curve for
Example 6-2



Step 2 We are looking for the area to the right of $z = -1.14$. This is an example of the second case. Look up the area for $z = -1.14$. It is 0.1271. Subtract it from 1.0000: $1.0000 - 0.1271 = 0.8729$. Hence, 87.29% of the area under the standard normal distribution curve is to the right of $z = -1.14$.

Note: In this situation, we subtract the area 0.1271 from 1.0000 because the table gives the area up to -1.14 . So, to get the area under the curve to the right of -1.14 , subtract the area 0.1271 from 1.0000, the total area under the standard normal distribution curve.

EXAMPLE 6-3

Find the area under the standard normal distribution curve between $z = 1.62$ and $z = -1.35$.

